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A Reference Dataset of 66 High-Quality Plastid Genomes across Three Algal Phyla

Jieqiong Wang, Kexin Liu, Xiaolei Wang, Zhengjie Yu, Chaoyong Wu, Shenglong Kan & Wei Zhang

Algae represent a diverse group of photosynthetic organisms, and their plastid genomes provide valuable molecular resources for investigating plastid evolution, photosynthetic adaptation, and phylogenetic relationships. However, algal plastomes exhibit remarkable evolutionary diversity, with substantial variation in genome structure and organization across different lineages. Currently, high-quality algal plastid genomes and a standardized and reliable workflow for generating high-quality algal plastid genomes is still lacking. To address these challenges, we successfully assembled 66 high-quality, complete algal plastid genomes, including 7 green, 19 brown, and 40 red algal genomes. During this process, we developed and established a standardized workflow for plastid genome assembly and quality assessment. Notably, these newly assembled genomes enrich existing NCBI resources by contributing 29 species, 13 genera, four families, and one order for the first time, offering both expanded taxonomic coverage and a standardized foundation for subsequent comparative and evolutionary analyses.

Background & Summary

Algae represent a major component of global primary production and play fundamental roles in marine carbon cycling and ecosystem functioning¹. Cyanobacteria are considered the common ancestor of extant green plant plastids, having formed distinct plastid structures in green and red algae through primary endosymbiosis². In contrast, brown algae underwent more complex secondary endosymbiotic events^{3,4}. The diverse evolutionary origins of plastids across algal lineages, together with their complex endosymbiotic histories, have shaped extensive variation in plastid genome structure, genome size, and gene content, thereby increasing the challenges of comparative analyses of plastid genome evolution across lineages^{5–10}. Currently, most algal plastid genomes have been generated through independent studies, resulting in heterogeneous assembly and annotation strategies and consequently variable data quality¹¹. Common issues include inconsistent genome annotations¹², unresolved structural errors, and inaccurate identification of inverted repeat (IR) regions, all of which reduce the consistency and reliability of available plastome resources.

Advances in sequencing technology have greatly accelerated research on plant chloroplast genomes. Although the numbers of algal plastid genomes have also increased, their expansion remains relatively limited compared with that of land plants. For instance, over 74,101 complete chloroplast genomes from Embryophyta have been publicly released. In comparison, the number of algal plastomes remains limited, with only 2,878 entries recorded as of November 2025, covering merely about 2% of algal species diversity¹³. Therefore, expanding high-quality algal plastid genome resources is essential for addressing the current gaps in available algal plastome data.

To address this gap, this study assembled 90 plastomes (66 complete circular genomes and 24 linear drafts) from 110 samples species, with 20 datasets excluded due to unresolved assembly graphs (Fig. 1A). These newly generated datasets add 29 species, 13 genera, four families, and one order previously absent from NCBI to the public repository (Fig. 1B). Together, these data establish an expanded plastome resource for downstream comparative genomic and evolutionary studies of algae.

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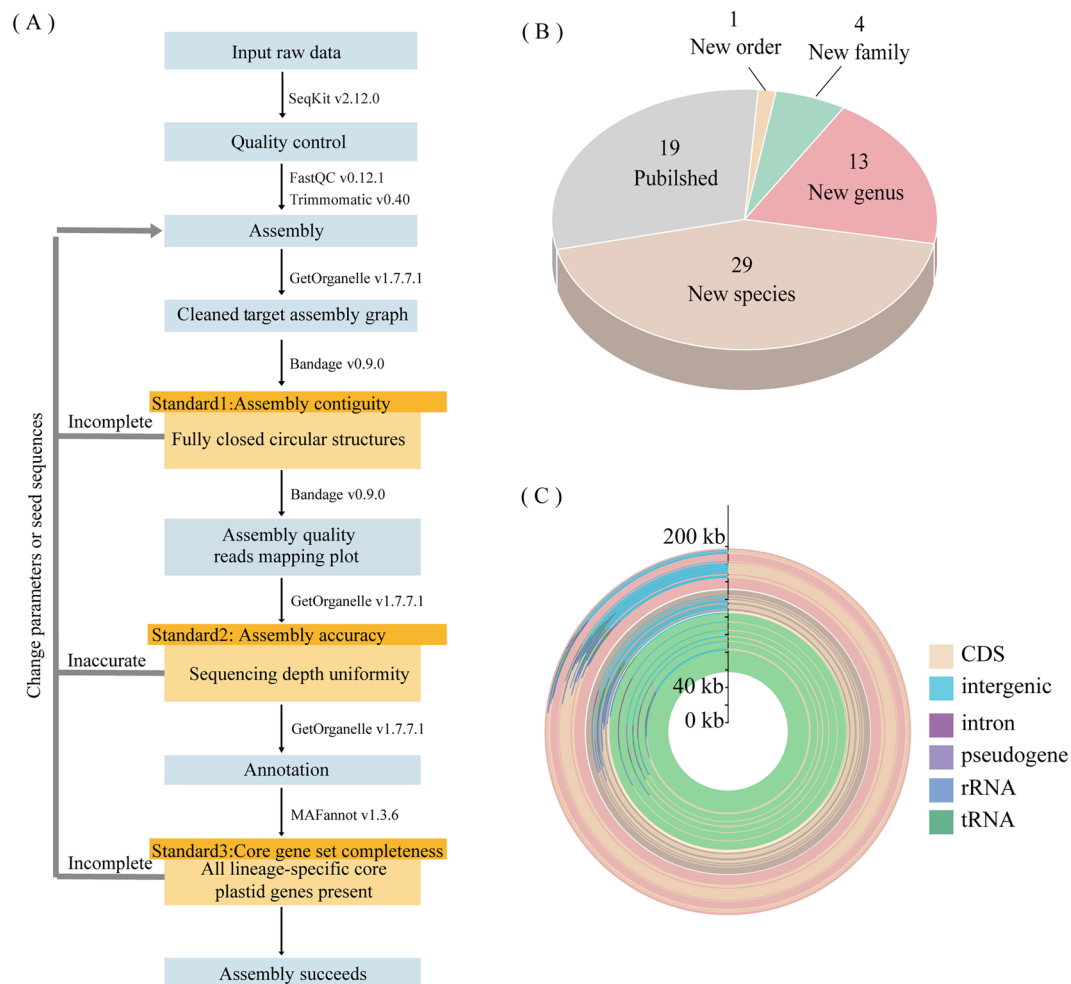


Fig. 1 Overview of algal genome assembly. **(A)** Schematic of the chloroplast genome assembly pipeline employing GetOrganelle. Key steps incorporating the three quality standards introduced in this study are annotated. **(B)** Taxonomic representation of the 66 high-quality assembled algal individuals. Taxonomic composition of the 66 algal individuals with high-quality genome assemblies. **(C)** Comparative analysis of genome size, structure, and gene content across investigated algae. Genome size is visualized as concentric rings: the outermost red ring corresponds to Rhodophyta, the middle brown ring to Ochrophyta, and the innermost green ring to Chlorophyta.

Methods

Data Sources. Raw sequencing reads were sourced from publicly available macroalgal genome datasets deposited in the NCBI Sequence Read Archive (SRA), including whole-genome sequencing data from 110 algal individuals and additional datasets reported in previous studies^{14–18}. These datasets correspond to BioProject accessions PRJNA924561, PRJDB15746, PRJEB39309, PRJNA32771, and PRJEB21674. Detailed information on species, source publications, BioProject accessions, and sequencing-run accessions is provided in Figshare repository at <https://doi.org/10.6084/m9.figshare.32983835.v8>¹⁹. To perform plastome assembly, annotation, and comparative analysis, we downloaded the publicly available Illumina short-read data for all taxa.

Plastomes Assembly and Annotation. Raw sequencing reads were quality-checked using FastQC v0.12.1.1 (<https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>) with default analytical settings and trimmed with Trimmomatic v0.39²⁰ to remove low-quality bases and adapter sequences with the following parameters: -phred33, ILLUMINACLIP:TruSeq. 3-PE.fa:2:30:10, HEADCROP:10, LEADING:20, TRAILING:24, CROP:140, SLIDINGWINDOW:4:15, AVGQUAL:28, MINLEN:36, and TOPHRED64. Cleaned reads were assembled with GetOrganelle v1.7.7.1 with -F other_pt, -R 10, and an appropriate plastid genome seed sequence specified using -s; other assembly parameters were left at their default settings. Assembly graphs were visualized using Bandage v0.8.1²¹ to inspect contig connectivity, coverage depth, and circularity, and were manually simplified based on depth and connectivity. Datasets were excluded when their assembly graphs could not be resolved into either a single uninterrupted circular plastome or a characteristic dumbbell-shaped structure associated with inverted repeats. Assembly quality was evaluated by remapping reads to the assembled plastomes using the evaluate_assembly_using_mapping.py script from GetOrganelle, which assesses per-site coverage, mismatch rates,

and indel patterns (Figure S1). The script was run using the corresponding paired-end reads (-1 and -2) and assembled plastome (-f), with mapping plots generated using—draw; other analytical parameters were left at their default settings.

Plastomes were annotated with MFannot v1.3.6 (<https://megasun.bch.umontreal.ca/apps/mfannot/>) using the bacterial/plastid genetic code (code 11)²². The resulting annotations in ASN format were converted to GenBank format using asn2gb from the NCBI toolkit²³. Annotations were manually curated, and circular genome maps were generated using OGDRAW v1.3.1²⁴ with default settings. In addition, key genomic features were manually compiled for 90 species.

Comparative plastomes analysis. Inverted repeat (IR) region loss, contraction, expansion, and boundary shifts were analyzed using IRplus²⁵ (irscope.shinyapps.io/IRplus/) with default settings and CPGView^{26,27} (<http://47.96.249.172:16085/cpgview/home>) with default settings (Figures S2–S4). Protein-coding genes were categorized by functional groups based on phylogenetic relationships, and their presence/absence patterns was recorded, with ORF-classified genes excluded from the analysis^{28,29}.

Phylogenetic Analyses. A maximum likelihood (ML) phylogenetic tree of algae was constructed using IQ-TREE multicore version 2.4.0³⁰ under the LG + F + I + G4 model, with *Synechocystis* sp. (CP007542) and *Prochlorococcus marinus* (BX548175) as outgroups. The procedure was as follows: (a) For all 92 species spanning three algal phyla, the nucleotide and corresponding protein sequences of every protein-coding gene were retrieved using the script `extract_all_genes_from_gb.py`. A set of 117 protein-coding genes present in at least 80% of species was selected and concatenated for tree construction. (b) Protein sequences of these genes were aligned using MAFFT³¹ with default parameters. (c) Unreliable alignment regions were removed using Gblocks with default parameters in PhyloSuite v1.2.3³². (d) PhyloSuite v1.2.3 was used to concatenate the 117 single-gene alignments into a supermatrix with default parameters. (e) IQ-TREE was applied to construct the ML tree under the LG + F + I + G4 model with 1,000 bootstrap replicates. (f) The tree was visualized and annotated using iTOL (<https://itol.embl.de/>)³³.

Data Records

The 66 assembled plastid genomes and their functional annotations are available under BioProject PRJEB121163 in DDBJ/ENA/GenBank databases, with individual listed in Table S2, and also provided via database links³⁴ and figshare (<https://doi.org/10.6084/m9.figshare.30943982.v4>)³⁵. Detailed plastid genome features, genome alignments, phylogenetic trees, and quality-control (QC) data are available at figshare: <https://doi.org/10.6084/m9.figshare.32983835.v8> (2026)¹⁹.

Data Overview

For chloroplast genome assembly, we obtained 66 complete plastomes and 24 linear draft assemblies, with circular maps of the complete genomes presented in Fig. 1C. The canonical quadripartite structure was displayed in all Ochrophyta plastid genomes, whereas it was observed only in a subset of Chlorophyta and Rhodophyta plastid genomes. In Rhodophyta, the repeat regions consisted of either inverted repeats or direct repeats (Fig. 2).

Among the Chlorophyta plastid genomes, large single-copy (LSC), small single-copy (SSC), and inverted repeat (IR) regions lengths ranged from 53,789–88,291 bp, 10,447–23,760 bp, and 316–5,266 bp, respectively. In contrast, Ochrophyta and Rhodophyta plastomes exhibited relatively conserved structural partitions within each lineage, with average LSC, SSC, and IR/repeat region lengths of 78.87/42.92/11.70 kb and 152.60/31.12/9.60 kb, respectively (Fig. 3A). The complete plastome length ranged from 81,997 to 113,888 bp in Chlorophyta and from 124,224 to 143,684 bp in Ochrophyta and from 164,327 to 194,981 bp in Rhodophyta (Fig. 3B). The average GC content varied among the three algal lineages, with mean values of 28.31% in Chlorophyta, 30.81% in Ochrophyta, and 29.98% in Rhodophyta (Fig. 3C). Correlation analysis showed that LSC length was significantly correlated with genome size (Fig. 3D), whereas SSC and IR lengths showed no significant correlations (Fig. 3E,F). Among the functional partitions, the protein-coding region (CDS) exhibited the strongest correlation with genome size, while tRNA and intergenic regions showed moderate correlations. In contrast, rRNA, introns, and pseudogenes showed little or no correlation (Fig. 3G–L).

Protein-coding gene numbers differed among the major algal lineages, averaging approximately 99 in Chlorophyta, 142 in Ochrophyta, and 186 in Rhodophyta. Compared with Ochrophyta and Rhodophyta, Chlorophyta exhibited reduced gene numbers across all functional categories, whereas the differences between Ochrophyta and Rhodophyta were primarily associated with genes involved in carbon assimilation and biosynthesis, transcription and translation, and light harvesting and chlorophyll biosynthesis (Fig. 4).

Technical Validation

Raw Data Quality Control. To ensure technical reliability, raw sequencing reads underwent quality control and trimming. All samples retained high Q30 scores, low error rates, and stable sequence quality.

Validation Criteria. *Standard 1–Assembly Contiguity.* Assembly graphs were visualized and evaluated using Bandage to assess their contiguity and circularity. A plastome was considered complete only when the final resolved graph formed one or more fully closed circular structures without gaps. Specifically, the graph was required to be continuous and fully resolved, without unconnected terminal nodes, dangling ends, open-ended paths, or fragmented structures with ambiguous connectivity.

Standard 2–Assembly Accuracy. Reads were remapped to the assembled plastomes to evaluate coverage. The sequencing-depth profile showed no abrupt or unexplained interruptions. Uniform depth distribution without significant fluctuations indicated correct and stable assembly.

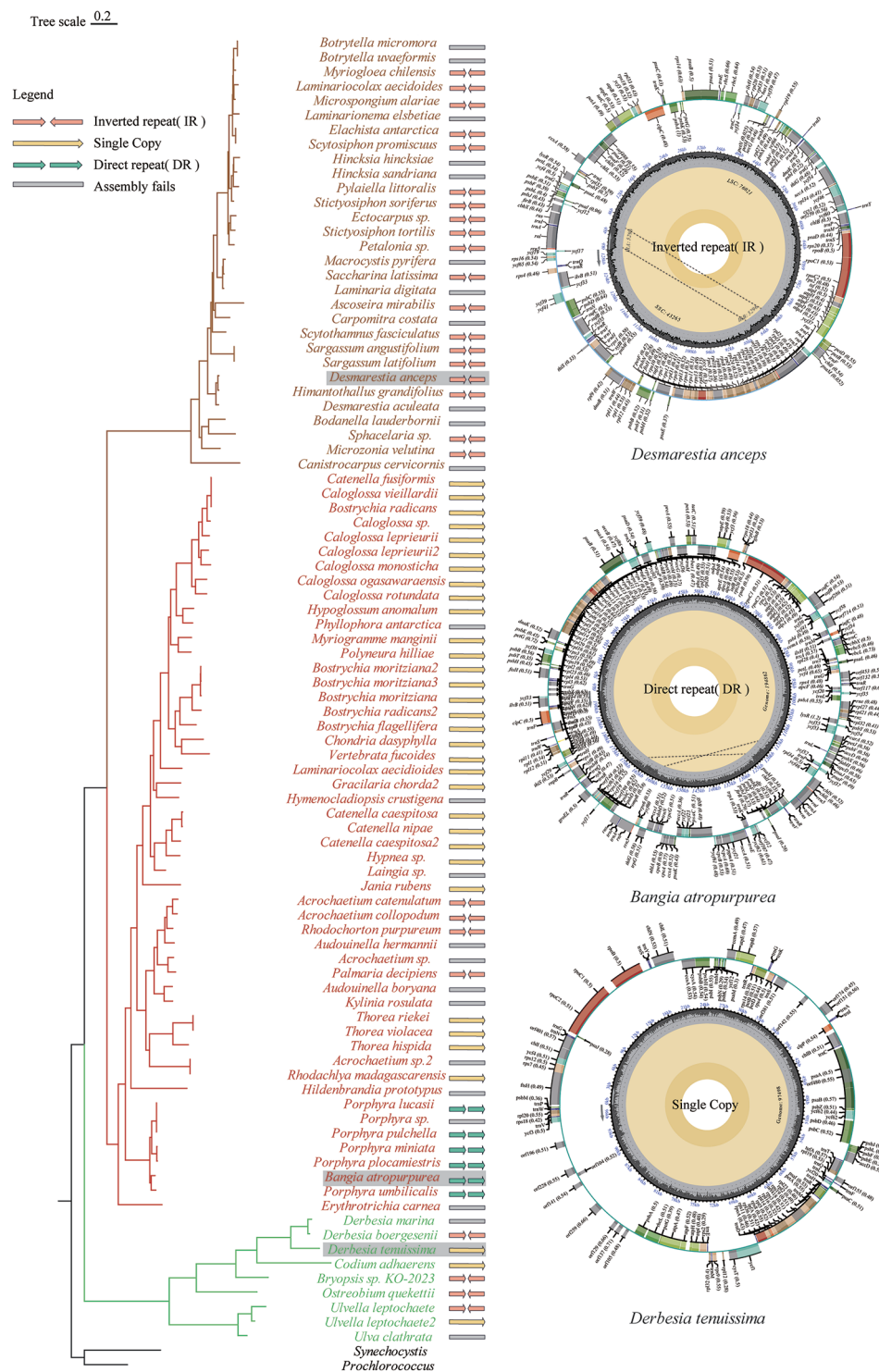


Fig. 2 The three primary chloroplast genome structures (inverted repeat, direct repeat, and single copy) and their distribution across 90 algal phylogeny.

Standard 3–Core Gene Set Completeness. Core gene set completeness was evaluated separately for Chlorophyta, Ochrophyta, and Rhodophyta, with the reference gene sets for each phylum detailed in Table S1. Plastome annotations retaining $\geq 80\%$ of the lineage-specific core genes were considered to have acceptable annotation completeness. Following automated annotation, all identified protein-coding genes, rRNAs, and tRNAs were manually curated against reference sequences from closely related species in GenBank.

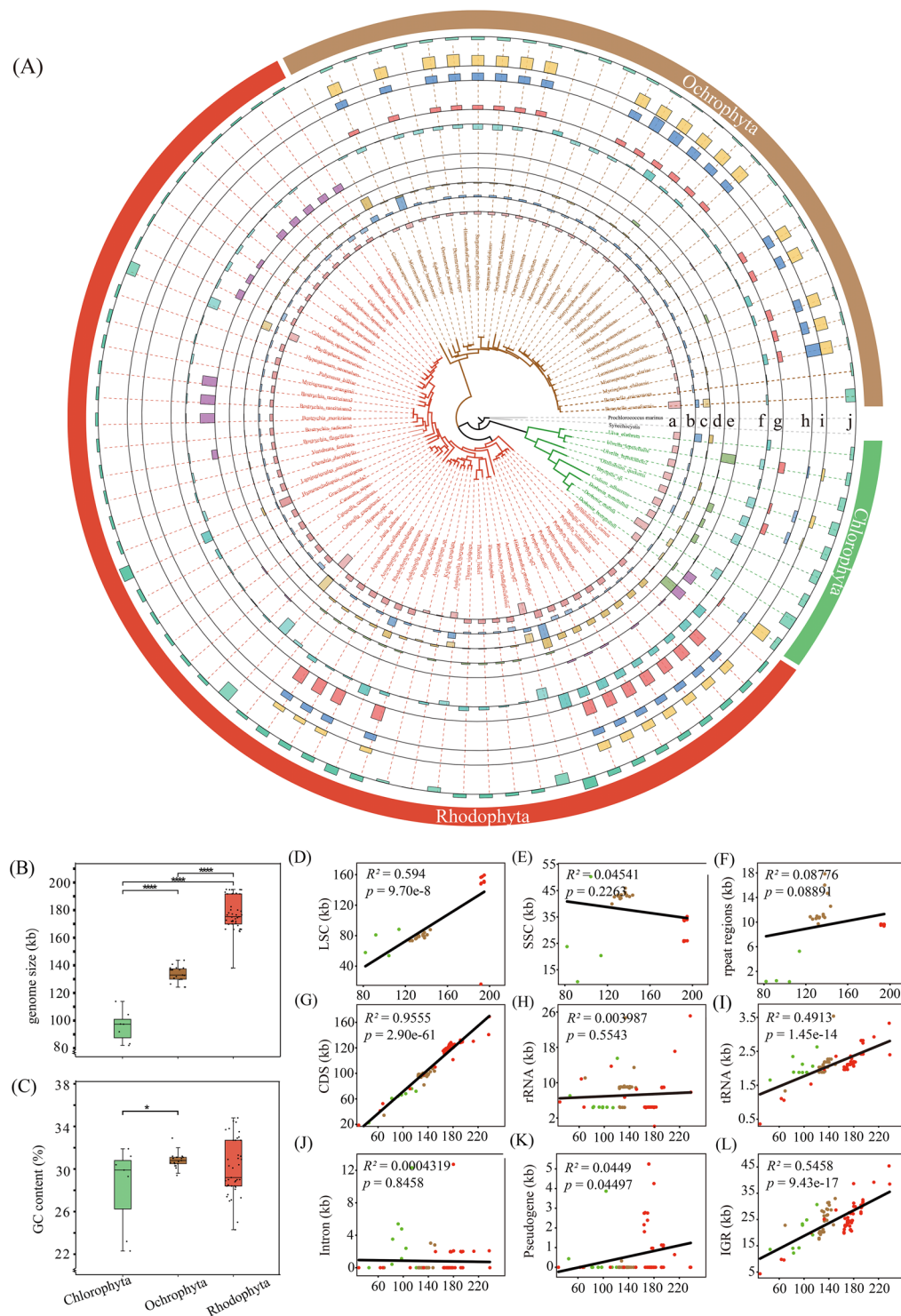


Fig. 3 Analysis of component sizes in algal chloroplast genomes and their correlation with genome size. **(A)** Summary statistics of plastid genome components across 90 algal phylogeny. The following components were quantified: (a) coding sequence (CDS) length, (b) ribosomal RNA (rRNA), (c) transfer RNA (tRNA), (d) introns, (e) pseudogenes, (f) intergenic regions (IGRs), (g) large single-copy (LSC) region, (h) small single-copy (SSC) region, (i) repeat regions, and (j) total genome size. In the radial plot, each ring corresponds to the mean value for a component. Values below the mean are indicated by inward-pointing bars, and values above the mean by outward-pointing bars. The bar height reflects the absolute deviation from the mean. **(B, C)** Boxplots comparing the distribution of total chloroplast genome size **(B)** and GC content **(C)** among the three algal phyla: Chlorophyta, Ochrophyta, and Rhodophyta. **(D–J)** Scatter plots depicting the correlations between total chloroplast genome size and the length of each major component: LSC **(D)**, SSC **(E)**, repeat regions **(F)**, CDS **(G)**, rRNA **(H)**, tRNA **(I)**, intron **(J)**, pseudogene **(K)** and IGR **(L)**.

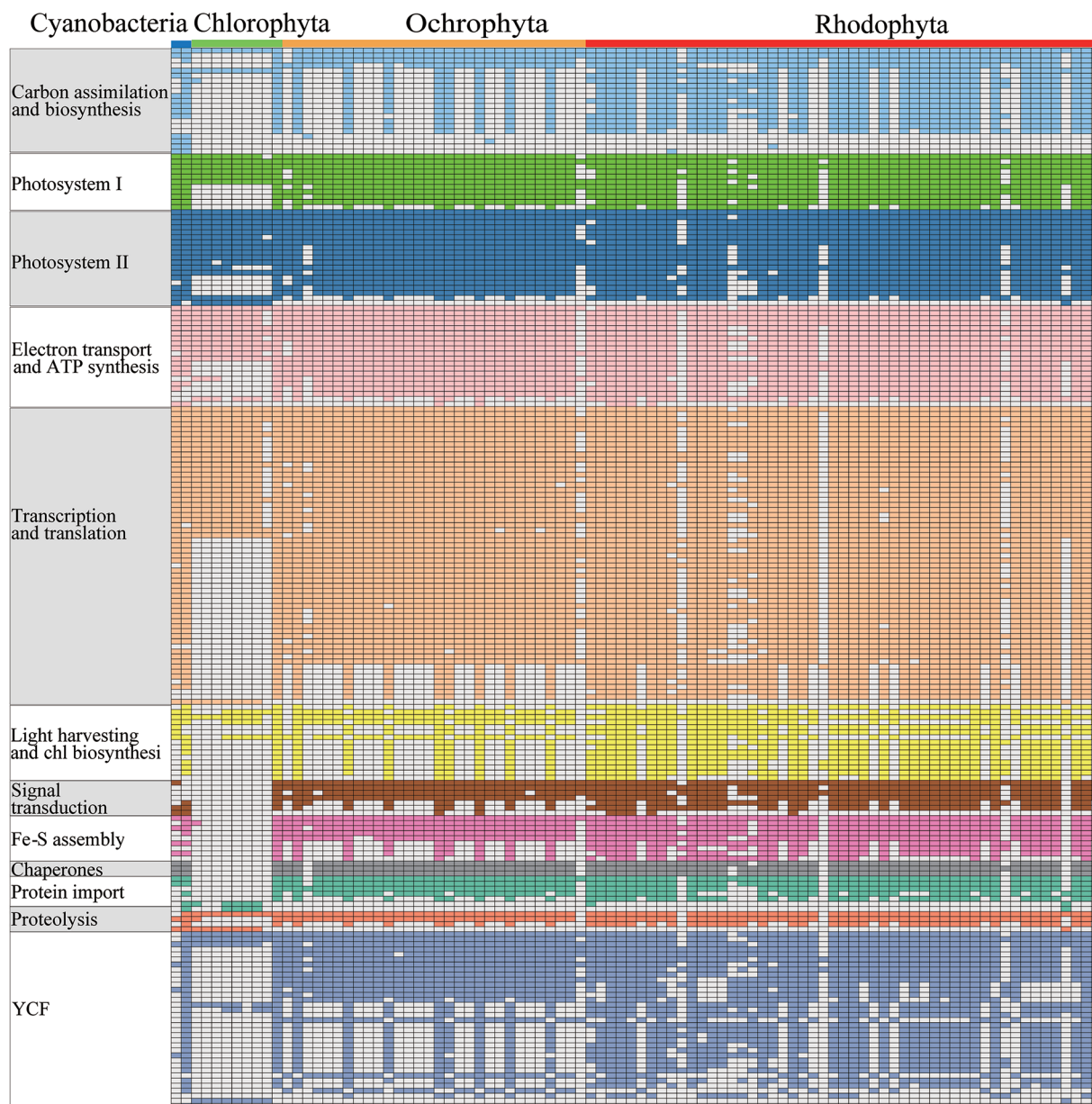


Fig. 4 Distribution of plastid-encoded protein-coding genes across major algal lineages. A presence/absence matrix for 90 representative species from Chlorophyta, Ochrophyta, and Rhodophyta. Colored ticks denote the presence of a gene in a given lineage.

Data availability

The 66 assembled plastid genomes and their functional annotations have been submitted to DDBJ/ENA/GenBank³⁴ and figshare (<https://doi.org/10.6084/m9.figshare.30943982.v4>)³⁵. All detailed plastome features, associated sequence data, including both raw and trimmed alignments, as well as the tree file are publicly available on figshare (<https://doi.org/10.6084/m9.figshare.32983835.v8>)¹⁹.

Code availability

The custom Python script `extract_all_genes_from_gb.py` used in this study is publicly available in figshare (<https://doi.org/10.6084/m9.figshare.30943982.v4>)³⁵ together with the plastid genome assemblies and annotation files.

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Author contributions

Jieqiong Wang: Data curation, Formal analysis, Methodology, Writing-original draft, Visualization. Kexin Liu: Data curation, Methodology. Xiaolei Wang: Conceptualization, Methodology. Zhengjie Yu: Data curation, Formal analysis. Chaoyong Wu: Conceptualization, Data curation. Shenglong Kan: Data curation, Methodology. Wei Zhang: Conceptualization, Data curation, Formal analysis, Methodology, Writing-review & editing.

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Competing interests

The authors declare no competing interests.

Additional information

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